SurgeSight

See a cyclone’s impact before it reaches the coast.

TEAM ECHO

Charan · Vishal

BUILD WITH AI: CODE FOR COMMUNITIES 2.0 · PROBLEM STATEMENT 05

THE PROBLEM

Forecasts tell us where a cyclone will go.
Not what it will hit.



Which school?

Which hospital?

Which road?

What if the track shifts?

THE CHALLENGE

Turn one cyclone forecast into answers every department can act on, in its own language, before landfall.

OUR SOLUTION

To solve it, we built SurgeSight. One bulletin in. Approved action out.

1 Read

Gemini

Gemini reads the IMD bulletin: text, PDF or photo

2 Score

Earth Engine

Every facility and main road scored for wind, surge and rain

3 Show

Google Maps

Alerted schools, hospitals and roads, each with its reason

4 Advise

Gemini

Each department's advisory, in its own language

5 Approve

Gmail · Firebase

An officer approves, then alerts go out

Numbers come from our models. Words come from Gemini.

CASE STUDY

Cyclone Michaung

December 2023 · Tamil Nadu and Andhra Pradesh

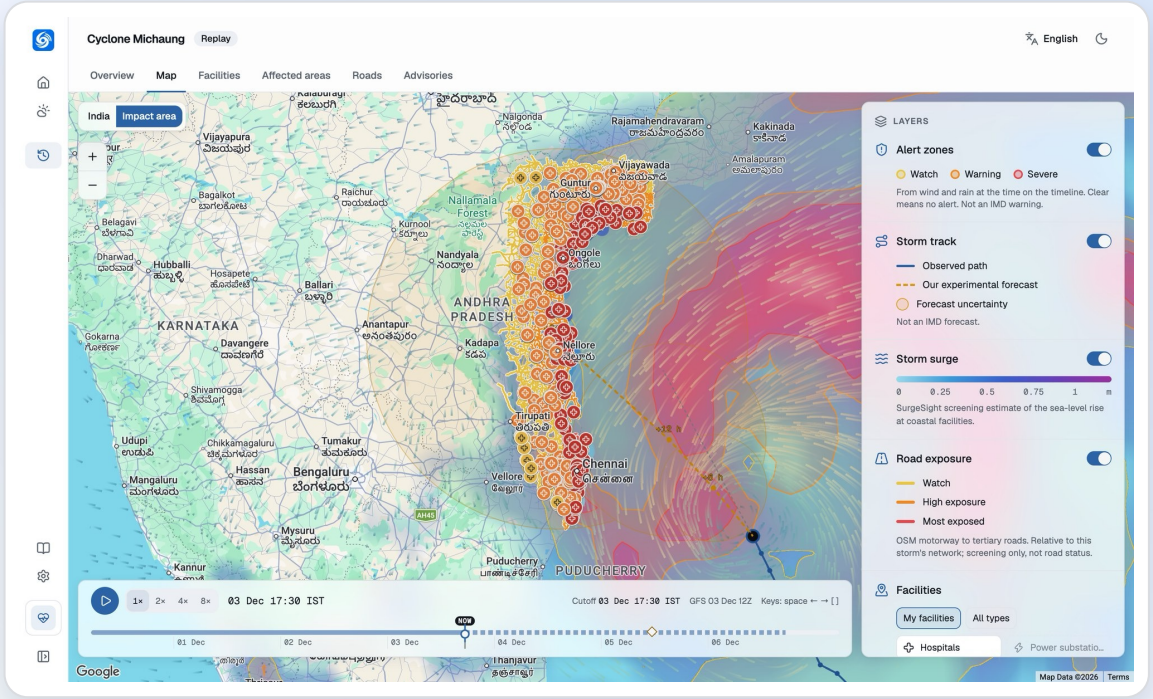
To study and test SurgeSight, we replayed it bulletin by bulletin, and checked our estimates against open satellite and weather data.

4,214
critical facilities

33,930 km
main roads

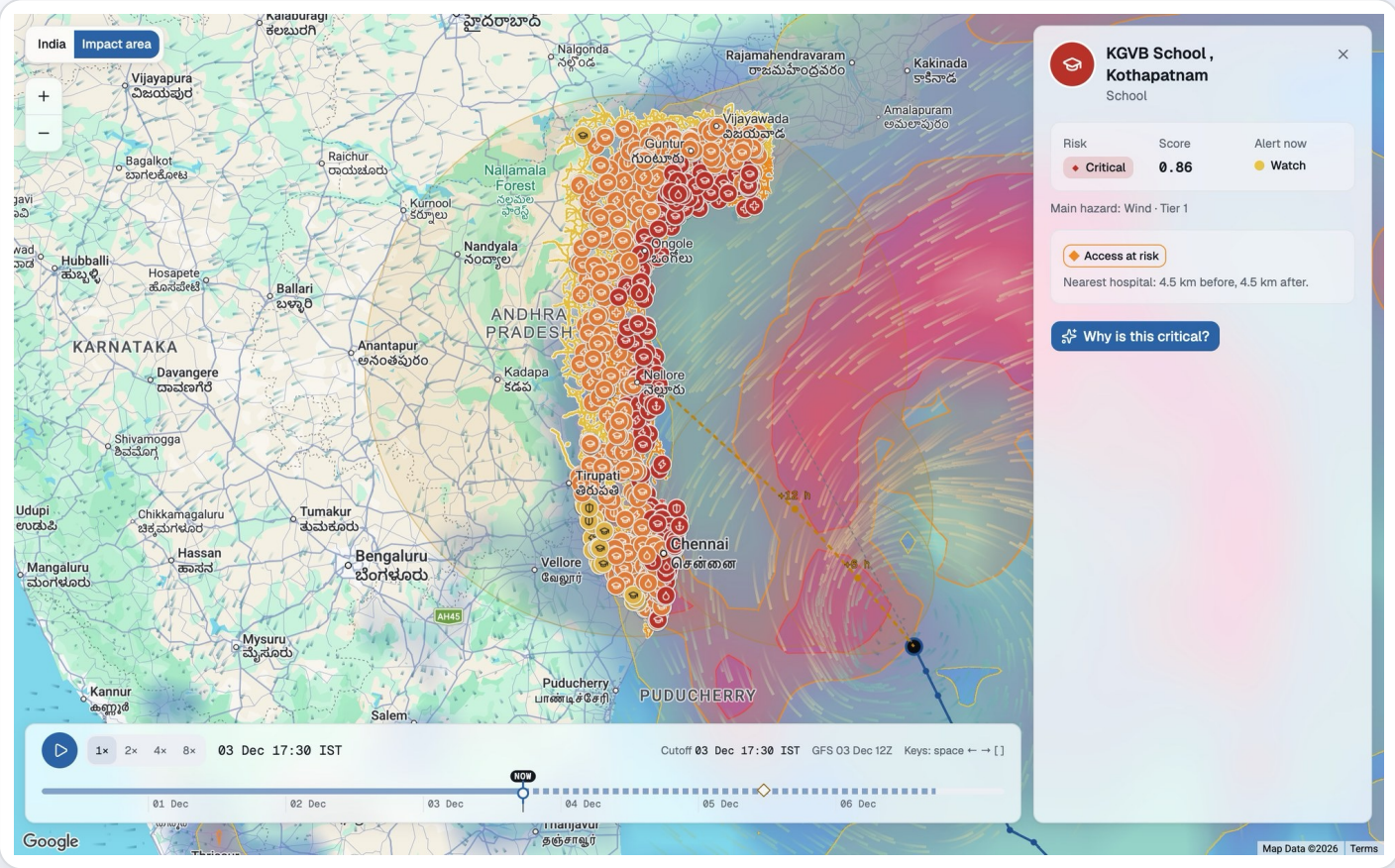
10.5 M
people in the priority area

7
departments



Two states, two languages · open data to check ourselves · an honest replay

Every school, hospital and substation under the storm, coloured by risk.



4,214 facilities, 8 types

Coloured from Low to Critical

Alert zones

Watch, warning and severe, moving with the storm

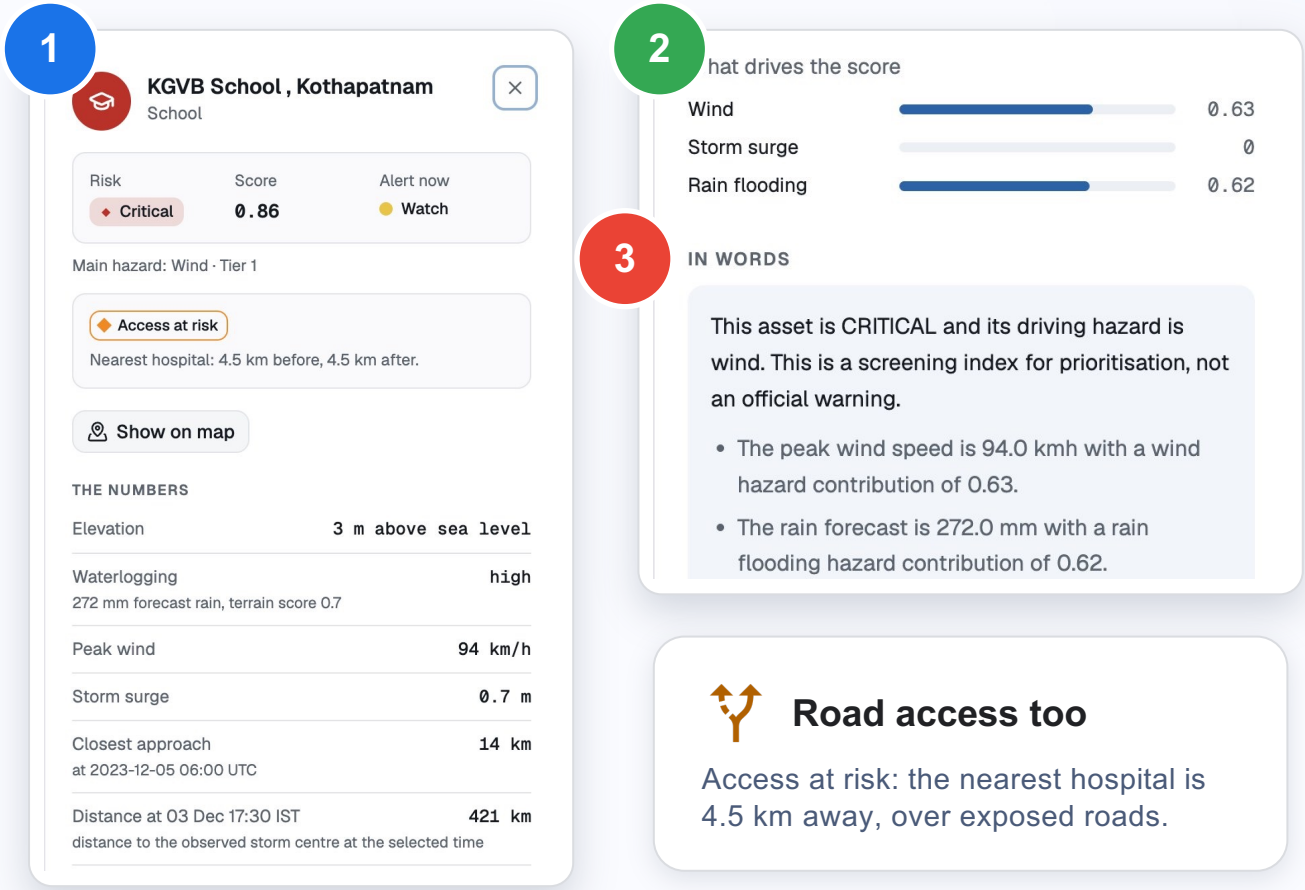
Track and forecast

Observed path, our forecast and its uncertainty

Roads to check

Main roads by exposure, and cut-off hospitals

Every alert comes with its reason: the numbers first, then Gemini’s words.



Road access too

Access at risk: the nearest hospital is 4.5 km away, over exposed roads.

1

The numbers

3 m above sea level, 272 mm of forecast rain, 94 km/h wind, 14 km from the track.

2

What drives it

Wind 0.63 and rain flooding 0.62 make this school Critical.

3

In words, by Gemini

Plain language for the officer. Every number is checked against the table.

What Gemini does in SurgeSight.

Reads

Turns an IMD bulletin (text, PDF or photo) into a storm track.

Explains

Why each school or hospital was alerted, and what a what-if changes.

Advises

An action plan for each of 7 departments, from the ranked facilities and roads.

Speaks six languages

English, Hindi, Tamil, Telugu, Bengali and Odia, with audio.

Reasons with tools

An ADK agent team picks from 29 tools to answer “what if?”

Searches

The latest bulletins and news with Google Search, with sources.

Gemini 3.7 Flash on Vertex AI. It never calculates the risk. A guard removes any number it cannot back up.

Each department gets its own advisory, in its own language.



CRITICAL Health Department

புயல் எச்சரிக்கை: சுகாதாரத்துறை மருத்துவமனைகளுக்கான அவசர இடர் தணிப்பு வழிகாட்டுதல்

மதிப்பிடப்பட்ட 2200 மருத்துவமனை அமைப்புகளில் 1070 அமைப்புகள் CRITICAL இடர் வகுப்பிலும் 1070 அமைப்புகள் HIGH இடர் வகுப்பிலும் அடையாளம் காணப்பட்டுள்ளன. மதிப்பீட்டின்படி 108 சுகாதார வசதிகள் துண்டிக்கப்பட்ட நிலையிலும் 1331 வசதிகளுக்கான அணுகல் இடரிலும் உள்ளன. சென்னை மற்றும் சுற்றியுள்ள முக்கிய சாலைகளில் 1696.7 கி.மீ மிக அதிக வெளிப்பாட்டு அபாயத்தில் (most exposed) உள்ளன.

அதிக இடர் எதிர்கொள்ளும் மருத்துவமனைகளில் மின்சார ஜெனரேட்டர் எரிபொருள், டயாலிசிஸ் மற்றும் தீவிர சிகிச்சைப் பிரிவு (ICU) தொடர்ச்சியை உடனடியாக உறுதி செய்யவும்.

முன்னுரிமைச் செயல்கள்

1. Punarjeevan Hospital, Dhanvantri Multi Speciality மற்றும் Dr.Kamakshi Memorial Hospital ஆகியவற்றில் ஜெனரேட்டர் எரிபொருள் இருப்பை உறுதிசெய்க.
2. துண்டிக்கப்பட்டதாகக் குறிப்பிடப்பட்டுள்ள Pulicat Government Hospital, RMS Ropheka Multi Speciality Hospitals மற்றும் Vasantha Hospital ஆகியவற்றில் அவசர நோயாளிகள் இடமாற்ற திட்டங்களைச் செயல்படுத்துக.
3. அணுகல் இடர் உள்ள Government Hospital, Area Hospital மற்றும் Sri Mundarapottamma

Gemini drafts
7 departments × 6 languages

Audio and a CAP 1.2 alert
Cloud Text-to-Speech, standard alert format

An officer approves
Nothing is sent automatically

Sent to the team
Gmail and phone push (Firebase)

HOW IT WORKS

Our models compute the risk. Gemini explains it. Both are checked.

DATA IN

IMD bulletin

Earth Engine · 10 datasets

OpenStreetMap

→

RISK MODELS

Wind · surge · rain

Roads · hospital access

Ensemble · what-if

→

GEMINI AGENTS

Gemini 3.7 Flash + ADK

29 tools

Number guard

→

OUT

Map with reasons

Six languages

Approved alerts

CHECKED AGAINST WHAT REALLY HAPPENED

27.6 km

average error of our 6-hour track forecast

0.71

match between forecast rain and NASA's observed rain

2×

satellites saw water twice as often on roads we flagged

23 Google services, each with a real job.

AI

- Gemini 3.7 Flash
- Vertex AI
- Gemini Live
- Agent Development Kit
- Search grounding
- Text-to-Speech
- Speech-to-Text

MAPS & EARTH

- Earth Engine
- Maps JavaScript
- Places
- Routes
- Geocoding
- Weather

DATA

- BigQuery GIS
- Firestore
- Cloud Storage

ALERTS

- Gmail API
- Firebase Messaging
- Firebase Auth

PLATFORM

- Cloud Run
- Cloud Scheduler
- Secret Manager
- Cloud Build
- Artifact Registry

Live on Cloud Run

- **One container**
  Scales to zero, scales out on a storm day
- **Refreshes itself**
  Cloud Scheduler job every 3 hours
- **Keys stay secret**
  Secret Manager, dedicated service account
- **New state = config file**
  Same app, models and agents

WHAT'S NEXT

From the control room to every coastal home.

A citizen app

"Am I at risk?" in my language, the nearest safe shelter and a lower-risk route.

Cascading failures

Which hospitals lose power or water when a substation floods.

Every phone

Approved alerts sent through NDMA's SACHET platform for cell broadcast.

Voice for villages

Officers call in and ask Gemini Live in Tamil, Telugu or Odia.

Money before landfall

Parametric triggers release relief funds before the storm hits.

Learns every storm

Satellite flood maps retune the model after each event.

WHY IT MATTERS

Forecasts into action, before landfall.

District Collectors
What to protect first, with the reason for each

Hospitals
Fuel, power and patient moves before roads flood

Power and water boards
Crews at the substations and pumps most at risk

People on the coast
Warnings in their own language

Scan to try SurgeSight

TEAM ECHO

Charan · Vishal

Live app

Demo video: [add link]

Code: [add link]